

INTERNSHIP PROJECT REPORT

IronWatch

Factory Walk-Through Security Gate for Ferrous-Metal
Theft Prevention with Facial-Recognition Attendance

Submitted by:	1. [Full Name] ID: [ETS XXXX/XX]
	2. [Full Name] ID: [ETS XXXX/XX]
	3. [Full Name] ID: [ETS XXXX/XX]
	(Add or remove rows to match the team size.)
Submitted to:	Department of Electromechanical Engineering, Addis Ababa Science and Technology University
Duration:	[Start Date] to [End Date]
Organization:	[Company Name] [Company Location]

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CERTIFICATION AND APPROVAL

This is to certify that the internship project entitled **“IronWatch — Factory Walk-Through Security Gate for Ferrous-Metal Theft Prevention with Facial-Recognition Attendance”** is a bona fide record of project work carried out by the student(s) listed on the cover page, in partial fulfilment of the requirements of the internship program of the Department of Electromechanical Engineering, Addis Ababa Science and Technology University (AASTU), and that the work was performed at the host organization stated on the cover page under joint academic and industrial supervision.

Role	Name	Signature	Date
Student 1	[Full Name]		
Student 2	[Full Name]		
Student 3	[Full Name]		
Company Supervisor	[Name, Title]		
Academic Advisor	[Name, Title]		
Head of Department	[Name, Title]		

Company stamp / seal:

DECLARATION

We, the undersigned, declare that this internship project report is our original work, that all sources of information and assistance have been duly acknowledged, and that the report has not been submitted, in whole or in part, for any other academic award. Simulation models, firmware, and documentation produced for this project are archived in the project repository **github.com/MINTESINOTESAYAS/Iron-Watch**.

Student 1 — [Full Name], ID: [ETS XXXX/XX]

Signature: _____ Date: _____

Student 2 — [Full Name], ID: [ETS XXXX/XX]

Signature: _____ Date: _____

Student 3 — [Full Name], ID: [ETS XXXX/XX]

Signature: _____ Date: _____

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ABSTRACT

Factories lose significant value to the unrecorded movement of ferrous materials — tools, fasteners, fixtures, and offcuts — through pedestrian exits, while paper attendance registers remain slow, error-prone, and vulnerable to buddy-punching. This report presents **IronWatch**, a low-cost factory walk-through security gate that combines **ferrous-metal theft prevention** with **facial-recognition attendance** in a single pedestrian lane.

The system comprises a 900 mm × 2000 mm × 350 mm steel gantry housing an LC search coil in one upright, a CD4046 phase-comparison channel with LM358 conditioning that collapses its DEMOD output toward 0 V for pure iron and steel while leaving small personal items such as keys and belt buckles above a firmware threshold (**$A_0 \leq 40 \Rightarrow \text{metal}$**), an Arduino Uno decision core driving red/green/buzzer signaling, a badge/face identity stage, and a **single pedestrian swing-arm boom barrier — the only moving part** — positioned by a MATLAB/Simulink-tuned **PID controller**. The boom arm was modeled from first principles (rotational dynamics, state-space form, and Jacobian linearization), sized mechanically (shaft, bearings, motor/gearbox, fabrication plan), and servo-tuned to a $0 \rightarrow 90^\circ$ step response with **0.68 s rise time, 1.18 s settling time, 1.5 % overshoot, and 0.01 % steady-state error** ($K_p=18$, $K_i=12$, $K_d=5$, filtered derivative, engage-zone integral with anti-windup). Firmware logic was verified in a Proteus bench in which a potentiometer stands in for DEMOD volts and a Virtual Terminal at 9600 baud stands in for the RFID badge reader (valid badge **A1B2C3**); all simulation substitutions are stated honestly, together with a commissioning plan for the physical gate. A company attendance log system — daily intern log, electronic gate register, and monthly summary — is specified in full.

Keywords: walk-through metal detection; ferrous discrimination; CD4046 phase comparison; Arduino; swing-arm boom barrier; PID control; MATLAB/Simulink; Proteus; facial-recognition attendance; mechatronics.

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LIST OF ABBREVIATIONS AND SYMBOLS

- **AASTU** — Addis Ababa Science and Technology University
- **ADC** — Analog-to-digital converter (10-bit, 0–1023 on Arduino Uno)
- **CSV** — Comma-separated values (attendance log export format)
- **DEMODO** — Demodulator output of the CD4046 phase comparator
- **DOF** — Degree of freedom
- **LQR** — Linear-quadratic regulator (considered and *not* selected — see §3.3.3)
- **PCB** — Printed circuit board
- **PID** — Proportional–integral–derivative (controller)
- **PLL** — Phase-locked loop (CD4046)
- **PWM** — Pulse-width modulation
- **RFID** — Radio-frequency identification (badge reader)
- **SHS** — Square hollow section (steel tube)
- **VCOIN** — VCO input, CD4046 pin 9 (must be biased — see §3.4.4)
- J, b, K_m, τ_c — Boom inertia, viscous damping, drive gain, Coulomb friction
- θ, ω, u — Boom angle, angular velocity, motor drive voltage

1 INTRODUCTION

1.1 Background of the Study

In discrete manufacturing and metalworking plants, ferrous materials are everywhere: hand tools, fasteners, jigs and fixtures, weld coupons, and saleable offcuts. Most of these items are small enough to be carried out unintentionally — or deliberately — through pedestrian exits, and conventional countermeasures (manual bag checks, security guards, CCTV review) are labor-intensive, inconsistent, and blind to *who* carried *what* through *which* gate at *what* time. Imported walk-through metal detectors solve only the sensing half of the problem: they are expensive to procure and maintain, they alarm on every key ring and belt buckle so that operators learn to ignore them, and they keep no attendance record of the screened workforce.

In parallel, many factories still run attendance on paper registers or isolated fingerprint terminals. Paper is slow to reconcile and easy to falsify; standalone terminals are disconnected from physical access, so an employee can be “present” in the register while product walks out of an unmonitored door. The natural engineering response is to **fuse screening, access control, and attendance into one pedestrian lane**: every passage is detected, identified, decided, and logged.

IronWatch is that fused lane, designed for local fabrication and maintenance. A steel walk-through gantry carries a ferrous-sensitive search coil; a CD4046 phase-comparison channel with LM358 conditioning converts the coil's phase response into a DC voltage that an Arduino Uno reads on A0; firmware discriminates bulk iron and steel from small personal items; a badge/face identity stage checks the pedestrian; and a single PID-controlled swing-arm boom — the only moving part — grants or denies passage while an attendance event is written to the company log. Models, firmware, and documentation are versioned in the public repository github.com/MINTESINOTESAYAS/Iron-Watch so the host company and future student teams can reproduce, audit, and extend the work.

1.2 Problem Statement

The host factory's pedestrian exits exhibited the following gaps, confirmed during the internship walk-throughs and supervisor interviews:

- **Unrecorded ferrous-material movement.** No sensing at pedestrian exits; tools, fasteners, and offcuts could leave the production area without any record or alarm.
- **No per-person gate record.** Even where a guard was posted, there was no reliable log linking a passage event to an individual identity.
- **Attendance disconnected from access.** The paper attendance register was reconciled manually, with errors, delays, and buddy-punching risk — and no cross-check against gate usage.
- **Imported-gate barrier.** Commercial walk-through detectors with acceptable false-alarm behavior were prohibitively expensive and required foreign spares and service.

- **False-alarm fatigue risk.** Any detector that cannot separate a steel bar from a key ring will be ignored by operators; discrimination is therefore a primary requirement, not a luxury.

The project therefore set measurable design targets: (i) a low-side analog trigger ($A0 \leq 40 \Rightarrow \text{metal}$) that fires on bulk iron/steel yet passes keys and buckles; (ii) a 90° boom swing completed in about one second with negligible overshoot; (iii) a deterministic grant/deny decision from detection $\times$ identity; and (iv) an attendance event logged for every passage decision.

1.3 Objectives

General objective. To design, model, simulate, and document a low-cost factory walk-through security gate that prevents ferrous-metal theft and records facial-recognition attendance, following a complete mechatronic workflow from mechanical sizing through control tuning to firmware verification.

Specific objectives.

- 1 Design the gantry frame and single swing-arm boom, and size the shaft, bearings, and motor/gearbox with documented safety margins and a fabrication plan.
- 2 Derive the boom-arm rotational dynamics, cast them into state-space form, linearize about the operating point, and verify controllability and observability.
- 3 Design and tune a MATLAB/Simulink **PID** angle servo (not LQR) for smooth, fast, overshoot-free open/close motion, with explicit anti-windup and saturation handling.
- 4 Design the electronics (LC coil + LM358 + CD4046 phase channel, Arduino Uno core, LED/buzzer signaling, badge stand-in, boom driver) and implement the threshold firmware.
- 5 Verify the firmware in a Proteus bench and the servo in MATLAB/Simulink, stating all simulation substitutions honestly and defining a physical commissioning plan.
- 6 Specify the company attendance log system (daily log, electronic gate register, monthly summary) linked to gate events.

1.4 Scope and Limitation

Scope. The project covers one single-lane pedestrian gate: a 900 mm clear opening for one person at a time; ferrous (iron/steel) detection only; a single-zone coil in one gantry upright; a single horizontal swing-arm boom as the only moving part; Arduino-based decision logic with serial badge/face-event input; PID boom-angle control; and simulation-level verification (Proteus + MATLAB/Simulink) with a defined bench commissioning procedure.

Limitations.

- No full electromagnetic field simulation of the coil-target interaction was performed; the analog front end is validated by circuit analysis, threshold design, and a bench calibration procedure (see §4.2).

- The facial-recognition pipeline is specified as an interface plus log format; in simulation it is represented by serial attendance events (badge string + result), not by a live camera model.
- Detection is single-zone and ferrous-only; non-ferrous metals (aluminum, copper) and multi-target localization are excluded.
- The gate is designed for indoor factory use; weatherproofing, tailgating optics, and turnstile-grade anti-passback are left as recommendations (§5.2).
- Gains tuned in simulation must be re-verified on the physical boom before production use.

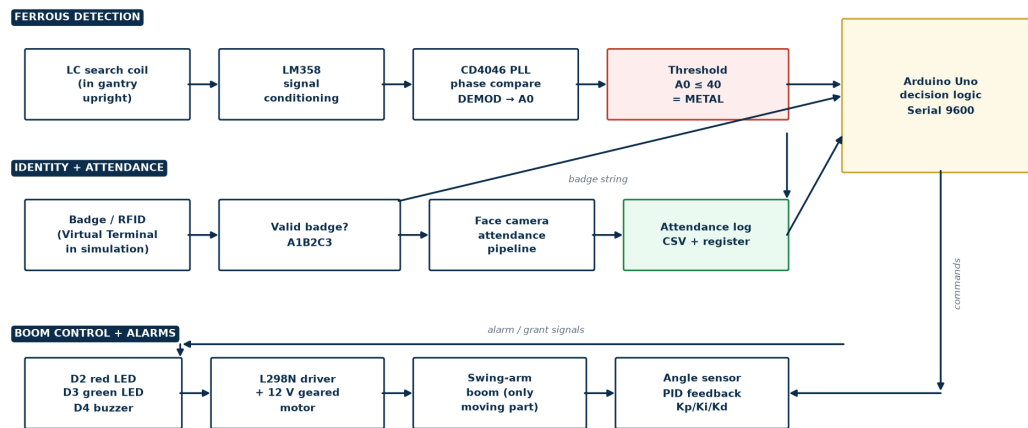
1.5 Methodology Overview

The work followed the mechatronic V-model in six steps:

- 1 **System analysis** — factory walk-throughs, exit-flow observation, and supervisor interviews to freeze requirements (opening width, throughput, alarm philosophy, log fields).
- 2 **Mechanical design** — gantry and boom geometry, load/inertia analysis, shaft, bearing, and motor/gearbox sizing, fabrication and assembly planning (§3.1).
- 3 **Mathematical modeling** — nonlinear rotational dynamics, state-space realization, and Jacobian linearization of the boom servo (§3.2).
- 4 **Control design and simulation** — open-loop analysis, controllability/observability, PID selection and tuning in MATLAB/Simulink with step-response and S-curve verification (§3.3, §4.1.2).
- 5 **Electrical/electronic design and firmware verification** — coil/LM358/CD4046 channel, Arduino firmware, and Proteus firmware-in-the-loop testing with declared substitutions (§3.4, §4.1.1).
- 6 **Attendance-log specification and reporting** — daily log, electronic register, and monthly summary formats validated against the outline requirements (Appendix C).

Figure 1 shows how the three subsystems — detection, identity/attendance, and boom control — meet at the Arduino decision core.

IronWatch system architecture — detection, identity, and boom control

**Figure 1.** System architecture — ferrous detection, identity/attendance, and boom control.

2 LITERATURE REVIEW

2.1 Walk-Through Metal Detection for Loss Prevention

Walk-through metal detectors (WTMDs) for personnel screening descend from airport-security technology, where multi-zone pulse-induction or continuous-wave portals localize metallic objects on the body (Franke, 2006). Industrial loss-prevention is a different optimization: the threat is bulk ferrous material rather than concealed weapons, throughput must match shift changes, and the total cost of ownership must suit a factory maintenance budget. Single-zone continuous-wave portals remain common in industry precisely because they are simple, robust, and cheap to keep running, at the price of coarse localization — an acceptable trade when the response is “stop and inspect” rather than “find the exact pocket” (Cumming, 2012). IronWatch follows this industrial lineage: one zone, one clear decision, one log entry.

A recurring lesson from deployment studies is that *operator trust* decides whether a portal keeps working after commissioning. Systems that alarm on every key ring train operators to wave people through, neutralizing the investment; discrimination — responding differently to threat-like and benign signatures — is therefore consistently identified as the difference between a security device and security theater (National Institute of Justice, 2003). The IronWatch threshold philosophy (bulk ferrous $\Rightarrow$ alarm; small personal items $\Rightarrow$ pass) is a direct response to this finding.

2.2 Phase-Sensitive Detection and Ferrous Discrimination

In a continuous-wave detector, a transmit coil excites the walk-through volume and a receive path observes the result. A metallic target changes both the amplitude and the *phase* of the received signal, and the phase carries composition information: high-permeability ferrous targets produce characteristically large phase shifts, while small or low-permeability objects produce small ones (Bruschini, 2002). Phase comparison against the excitation reference therefore separates “big iron” from “small clutter” far better than amplitude alone — the physical basis of IronWatch's CD4046 channel.

The CD4046B phase-locked loop (Texas Instruments, 2018) is a convenient vehicle for this idea: its phase comparators convert a phase difference into a demodulated DC voltage (DEM0D) that a microcontroller can threshold directly, without DSP. The LM358 dual op-amp (STMicroelectronics, 2021) provides the accompanying gain, filtering, and buffering with single-supply operation suited to factory-floor builds. A well-known commissioning pitfall — emphasized in the IronWatch design — is that the 4046's VCO input (pin 9, VCOIN) must be properly biased; left floating, the VCO stalls and DEMOD parks at a rail, masquerading as a permanent detection (see §3.4.4).

2.3 Pedestrian Barriers and Boom-Gate Mechanisms

Pedestrian access hardware spans tripod turnstiles, swing gates, sliding leaves, and boom arms. Swing-arm booms are favored where simplicity and visibility matter: one pivot, one actuator, an unambiguous open/closed state, and easy breakaway or manual override for emergencies (North American Barrier Association, 2019). Their dynamics are those of a single rigid link about a fixed axis — inertia, viscous and Coulomb friction, and (for vertical-lift variants) gravity — which makes them a textbook single-input single-output servo problem (Nise, 2020).

Machine-element sizing for such light mechanisms follows standard practice: torsion design of the pivot shaft, L_{10} rating-life selection of rolling bearings, and torque/speed matching of a geared motor to a specified motion profile with an explicit service factor for intermittent duty (Khurmi & Gupta, 2005; Budynas & Nisbett, 2020). IronWatch applies each of these steps in §3.1 and deliberately keeps the mechanism to a single moving part to minimize wear points, pinch hazards, and maintenance skill requirements.

2.4 Biometric Attendance and Face Recognition in Industry

Biometric attendance — fingerprint, face, or iris — addresses the central weakness of cards and registers: the credential cannot be lent to a colleague (Jain et al., 2011). Face recognition is the least intrusive option for a walk-through lane because it needs no contact and no pause if the camera is positioned on the approach; modern embedding-based matchers achieve high accuracy under controlled indoor lighting (Schroff et al., 2015). Deployment literature stresses three complements that IronWatch adopts as requirements: enrollment under supervision (HR office), liveness/anti-spoofing before any access grant, and a tamper-evident log that binds each identity event to a timestamp, gate, and decision (ISO/IEC 30107-1:2023). In the IronWatch prototype the camera pipeline is represented by its serial-event interface and log format, so the gate logic and attendance schema can be verified now and the vision module integrated later without changing the decision core.

2.5 Model-Based Design and Simulation in Mechatronics

Model-based design — deriving control from a dynamic model, validating in simulation, then deploying to embedded hardware — is standard mechatronic practice (Ogata, 2010; MathWorks, 2024). For microcontroller projects, two benches dominate: MATLAB/Simulink for control-loop tuning against a plant model, and Proteus for firmware-in-the-loop checks of the actual compiled code against virtual peripherals (Labcenter Electronics, 2023). Used honestly, with every substitution declared (a potentiometer for a sensor voltage, a terminal for a reader, an ideal motor for a geared drive), these benches catch the great majority of logic and tuning defects before metal is cut — which is exactly how IronWatch uses them in Chapter 4.

3 SYSTEM DESIGN

3.1 Mechanical Design

The mechanical system has exactly one moving part — the swing-arm boom. Everything else (gantry, coil housing, drive post, electronics cabinet) is static structure. This section sizes the structure, the pivot, and the drive for a 90° pedestrian swing with a smooth S-curve profile, and closes with the fabrication and assembly plan.

Gantry frame — 900 mm walk-through × 2000 mm high × 350 mm deep (SHS 40×40×2)

Front view (walk-through)

Side view (one upright + boom)

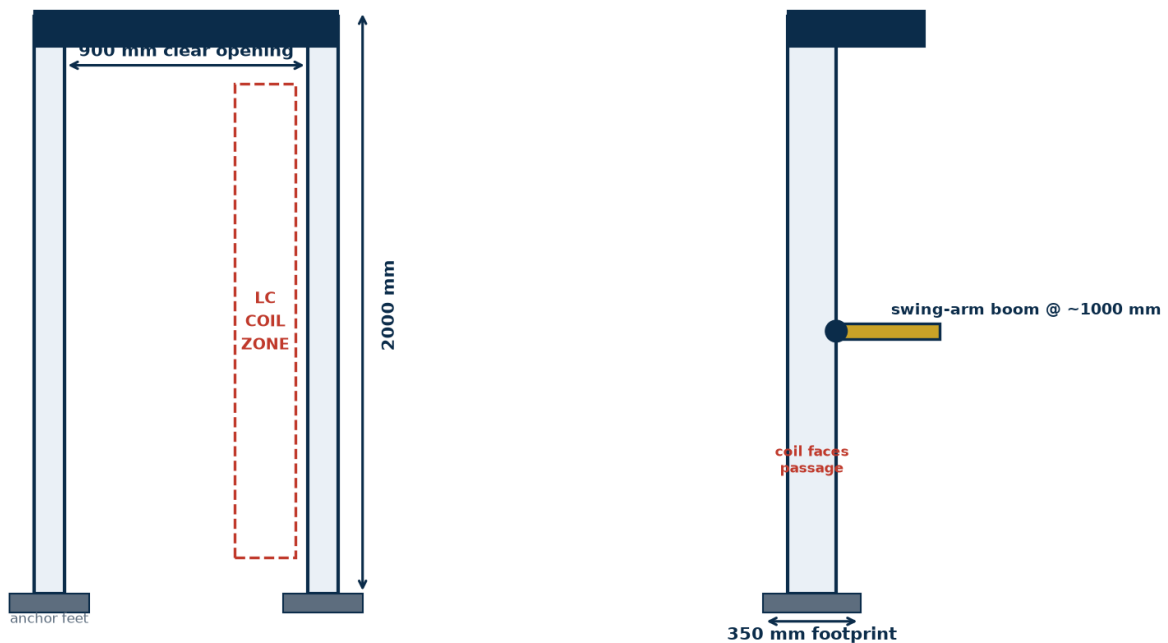


Figure 2. Gantry frame — 900 mm walk-through × 2000 mm high × 350 mm deep (SHS 40×40×2).

3.1.1 Load Analysis

The gantry is a portal frame in 40 × 40 × 2 mm square hollow section (SHS), giving a 900 mm clear walk-through, 2000 mm overall height, and a 350 mm deep foot for freestanding stability (Figure 2). Static loads are the frame self-weight (≈ 28 kg with header and feet), the coil panel (≈ 4 kg) in the right upright, the drive post with motor and boom (≈ 9 kg), and the electronics cabinet (≈ 5 kg). The governing accidental load is a 150 N horizontal push at boom-tip height (1.0 m) from a pedestrian leaning on the closed arm; reacted at the 350 mm-spaced anchor feet, it produces a restoring-moment demand far below the frame's self-righting moment once the feet are anchor-bolted (M8 expansion anchors, four per foot).

The boom itself is a 0.85 m aluminum tube (1.2 kg) with a 0.3 kg paddle/end-stop assembly at the tip. Its mass moment of inertia about the vertical pivot, including hub and shaft, is $J_{\text{arm}} \approx 0.53 \text{ kg}\cdot\text{m}^2$; adding the gearbox-reflected rotor inertia gives the control-plant value $J = 0.59 \text{ kg}\cdot\text{m}^2$ used throughout §3.2–§3.3. Bearing radial load is dominated by the overhung boom weight ($\approx 15 \text{ N}$) plus drive-post preload, conservatively taken as $P = 60 \text{ N}$ for life calculations.

3.1.2 Design of Machine Elements

Frame joints are MIG-welded all round, with M8 bolted foot plates for leveling and anchoring. The coil zone is a **non-metallic** window (FR-4 / polycarbonate panel) set into the right upright so the steel frame does not short the search field; the coil is potted in epoxy against vibration and humidity. The boom pivot runs in two 6004 deep-groove ball bearings housed in a welded drive post (see §3.1.4). Travel is bounded by rubber end-stops at 0° and 90° plus electrical limit switches; the stops absorb the residual $\approx 1^\circ$ settling tail of the servo (see §4.1.2) so the arm always seats firmly without hammering. All fasteners are property class 8.8, and pinch points at the pivot are covered by a sheet-metal guard.

3.1.3 Boom-Arm and Drive Geometry

Figure 3 shows the boom in plan view: closed, the 0.85 m arm spans the 900 mm passage at 1000 mm height; open, it swings 90° to lie along the exit side. A 12 V, 80 W DC gearmotor (40:1, rated 6 N·m continuous, 7.8 N·m stall) drives the Ø20 mm pivot shaft directly through a keyed hub — no belt, no chain, no extra bearings. Required torque follows the servo sizing equation:

$$\tau_{req} = J\alpha + b\omega + \tau_c, P_{mech} = \frac{\tau\omega}{\eta} \quad (3.1)$$

With $J = 0.59 \text{ kg}\cdot\text{m}^2$, $b = 1.20 \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$ (bearings, gearbox, and reflected back-EMF damping), Coulomb friction $\tau_c = 0.25 \text{ N}\cdot\text{m}$, and the S-curve profile peaks $\alpha = 2.5 \text{ rad}/\text{s}^2$, $\omega = 1.8 \text{ rad}/\text{s}$, the demand is $\tau_{req} \approx 3.88 \text{ N}\cdot\text{m}$, giving a continuous service factor of ≈ 1.5 — standard for intermittent (S3-25 %) gate duty — and a transient/stall margin of ≈ 2.0 . Mechanical drive power is $\approx 15 \text{ W}$, comfortably inside the 80 W motor frame.

Swing-arm boom — plan view, 90° travel (the single moving part)

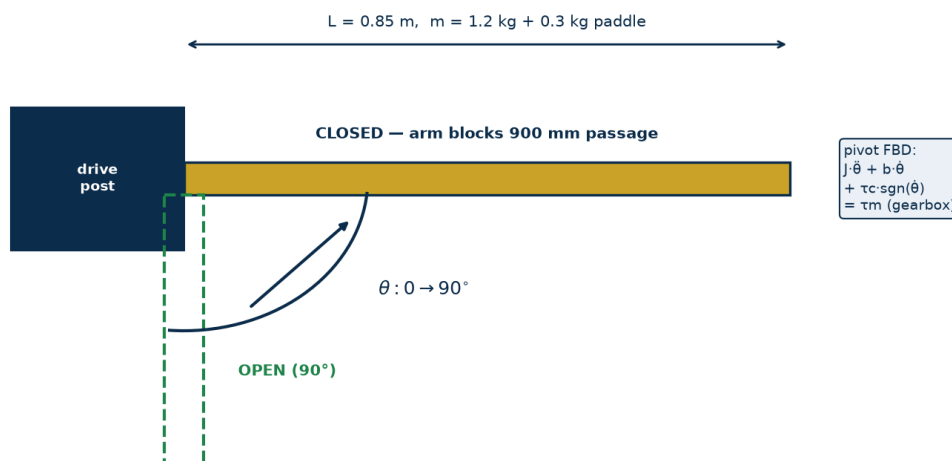


Figure 3. Swing-arm boom — plan view and 90° travel (the single moving part).

3.1.4 Bearing Design

Two 6004 deep-groove ball bearings (20 × 42 × 12 mm, dynamic rating $C = 9.4$ kN) support the pivot shaft in a spaced housing that reacts overturning moment as a force couple. Rating life follows:

$$L_{10} = (C/P)^3 \times 10^6 \text{ rev} \quad (3.2)$$

At the conservative equivalent load $P = 60$ N, L_{10} exceeds 10^{12} revolutions ($\approx 10^9$ h at gate duty) — effectively infinite, meaning the bearings are selected for bore size, stiffness, and sealing (2RS dust covers for the factory environment) rather than fatigue life. The housing bores are machined in one setup for coaxiality.

3.1.5 Shaft Design

The pivot shaft is Ø20 mm C45 steel, keyed to the boom hub (6 × 6 mm parallel key) and to the gearbox output. Torsional shear under the worst case — a stalled motor at 7.8 N·m — follows:

$$\tau_{max} = 16T/(pi d^3) < = \tau_{allow} \quad (3.3)$$

giving $\tau_{max} \approx 5.0$ MPa against an allowable ≈ 55 MPa: a large margin that also covers shock from pedestrians striking the arm. Bending from the overhung boom (≈ 13 N at 0.45 m mean radius) adds negligible combined stress. A circlip and shoulder locate the shaft axially; a Woodruff-key alternative was rejected to keep spares common.

3.1.6 Fabrication and Assembly

The build sequence is specified for a basic factory workshop (cut-off saw, MIG welder, pillar drill, angle grinder, paint booth):

- 1 **Cut and prep** — SHS members cut to length with 45° miters at the header corners; deburr and degrease all parts.
- 2 **Weld** — Tack, check square and diagonals (opening 900 ± 2 mm), then weld all round; weld foot plates and the drive-post housing.
- 3 **Machine** — Bore the bearing housing in one setup; drill anchor, switch-bracket, and cabinet holes; ream the boom-hub keyway.
- 4 **Coil panel** — Wind and pot the LC search coil, fit it into the non-metallic upright window, and route its leads in conduit to the cabinet.
- 5 **Finish** — Prime and enamel (safety-yellow boom with black chevrons; grey frame); fit rubber end-stops, guards, and limit switches.
- 6 **Assemble and align** — Set bearings with light preload, check the boom sweeps 0–90° without rubbing, anchor the feet, torque M8 anchors to 20 N·m, and run the 500-cycle endurance check of §4.2.

Table 1. Mechanical parameters and sizing results.

Parameter	Value	Remarks
Clear opening × height × foot depth	900 × 2000 × 350 mm	Figure 2; SHS 40×40×2
Boom length / mass	0.85 m / 1.2 + 0.3 kg	Al tube + paddle/stop
Pivot height / travel	≈ 1000 mm / 90°	Single moving part
Boom inertia J (plant)	0.59 kg·m ²	Arm + hub + reflected rotor
Damping b / Coulomb τ _c	1.20 N·m·s/rad / 0.25 N·m	Incl. back-EMF term
Torque demand τ _{req}	≈ 3.88 N·m	S-curve peaks α=2.5, ω=1.8
Motor / gearbox	12 V 80 W DC / 40:1	Rated 6 N·m, stall 7.8 N·m
Service factor	≈ 1.5 cont. / ≈ 2.0 stall	Intermittent S3-25% duty
Shaft	Ø20 mm C45, 6×6 key	τ _{max} ≈ 5 MPa □ 55 MPa
Bearings	2× 6004-2RS	L10 effectively infinite
Drive power (mech.)	≈ 15 W	12 V supply, L298N driver

3.2 Mathematical Modeling

The plant for control design is the boom pivot: motor voltage in, boom angle out. Detection logic (threshold, badge, alarms) only *triggers* reference changes — it is not part of the servo plant, a separation enforced throughout §3.3.

3.2.1 Rotational Dynamics of the Swing-Arm Boom

Applying Newton's second law for rotation about the fixed vertical pivot, with motor torque τ_m driving inertia J against viscous damping b, gravity, and Coulomb friction τ_c , gives the nonlinear equation of motion:

$$J\ddot{\theta} + b\dot{\theta} + mgl_c \sin \theta + \tau_c \text{sgn}(\dot{\theta}) = \tau_m \quad (3.4)$$

where θ is the boom angle from closed ($0 \rightarrow \pi/2$ rad), m the arm mass, $l_c \approx 0.42$ m the center-of-mass radius, and $\text{sgn}(\cdot)$ the signum function. Two simplifications apply to IronWatch: (i) the arm swings about a **vertical** axis, so gravity does no work on the motion and the $m \cdot g \cdot l_c \cdot \sin \theta$ term vanishes (it is retained in the derivation so the model also covers lift-type arms); and (ii) the geared DC drive is represented by its electrical and torque relations:

$$L \frac{di}{dt} + Ri + K_b \omega_m = v, \tau_m = NK_t i \quad (3.5)$$

with armature inductance L, resistance R, back-EMF constant K_b , torque constant K_t , motor speed ω_m , terminal voltage v, and gear ratio N. Because the electrical time constant ($L/R \approx$ a few ms) is two orders of magnitude faster than the mechanical response, the current loop is collapsed into the static drive gain $K_m = 0.65$ N·m/V at the

output shaft, with back-EMF absorbed into the viscous coefficient b — the standard servo reduction (Ogata, 2010).

3.2.2 Linear State-Space Representation

Choosing the state $x = [\theta, \omega]^T$ with $\omega = d\theta/dt$, input $u = v$ (volts, $|u| \leq 12$ V), and output $y = \theta$, and dropping the gravity term (horizontal swing) and the non-smooth Coulomb term for the linear design model (both are reintroduced for verification in §4.1.2), yields:

$$\dot{x} = A \cdot x + B \cdot u, \quad y = C \cdot x + D \cdot u \quad (3.6)$$

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -b/J \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ K_m/J \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0 \quad (3.7)$$

Numerically, with $J = 0.59$ and $b = 1.20$, $A = [0, 1; 0, -2.03]$ and $B = [0; 1.10]$. The corresponding open-loop transfer function is:

$$\Theta(s)/U(s) = K_m / [s \cdot (J \cdot s + b)] = 0.54 / [s \cdot (0.49 \cdot s + 1)] \quad (3.8)$$

a type-1 servo with mechanical time constant $\tau = J/b \approx 0.49$ s: a voltage step produces a ramping angle, so feedback is mandatory for positioning.

3.2.3 Linearization Using Taylor Expansion and Jacobian

Formally, the nonlinear vector field $f(x, u)$ from §3.2.1 is linearized about an operating point (x_0, u_0) by first-order Taylor expansion, whose coefficient matrices are the Jacobians:

$$A = df/dx|_{x_0, u_0}, \quad B = df/du|_{x_0, u_0} \quad (3.9)$$

Evaluating $\partial f/\partial x$ and $\partial f/\partial u$ at the closed-rest point $(\theta_0 = 0, \omega_0 = 0, u_0 = 0)$ gives $A_{11} = 0$, $A_{12} = 1$, $A_{21} = -m \cdot g \cdot l_c \cdot \cos\theta_0/J = 0$ (horizontal swing; for a lift arm this would be $-m \cdot g \cdot l_c/J$), $A_{22} = -b/J$, $B = [0; K_m/J]$ — confirming the matrices of §3.2.2 as the exact Jacobian linearization. The smooth Coulomb approximation $\tau_c \cdot \tanh(\omega/0.05)$ used in simulation has zero slope contribution at rest speed in the design model and is likewise reintroduced only at verification.

3.3 Controller Design

3.3.1 Open-Loop Response Analysis

The open-loop plant is a velocity-damped integrator: it cannot hold an angle, drifts under any disturbance torque (pedestrian contact, seal drag), and its step-voltage response is an exponential approach to a steady *speed* ($\omega_{ss} = K_m \cdot u/b \approx 0.54 \cdot u$ rad/s) — i.e., an angle ramp that would slam the boom into its end-stop. Poles at $s = 0$ and $s = -b/J = -2.03$ confirm marginal stability with no position stiffness. Closing a position loop with integral action is therefore required both for set-point tracking and for disturbance rejection at the seated positions.

3.3.2 Controllability and Observability

For the second-order realization, the Kalman controllability matrix $Q_c = [B, A \cdot B]$ evaluates to $[[0, 1.10], [1.10, -2.24]]$ with determinant $-1.21 \neq 0$, so $\text{rank}(Q_c) = 2 = n$: the boom angle and velocity are fully controllable from the motor voltage. The observability matrix $Q_o = [C; C \cdot A] = [[1, 0], [0, 1]]$ is the identity, so $\text{rank}(Q_o) = 2$: angle measurement alone (potentiometer or encoder) observes the full state with no observer required. These checks license the simple output-feedback PID of §3.3.4.

3.3.3 Control Strategy Selection

The outline template suggested LQR; this project deliberately selects **PID** instead, for reasons recorded here so the deviation is auditable:

- **Plant order and structure.** A second-order, minimum-phase, fully observable SISO servo needs no optimal state feedback; PID places the dominant poles just as effectively with far fewer assumptions.
- **Embedded reality.** The production controller is an Arduino Uno. A filtered PID with clamped integration runs in a dozen lines of fixed-point-friendly code; LQR needs a gain solve, full-state access, and re-tuning tooling the factory does not have.
- **Field service.** Factory technicians can understand and re-tune three gains with a stopwatch; Q/R weighting matrices are opaque to them. Maintainability is a design requirement.
- **Constraint handling.** The dominant nonlinearities are saturation (± 12 V), Coulomb friction, and end-stops — handled transparently by PID anti-windup plus an S-curve reference, whereas LQR would need additional governors anyway.
- **Performance achieved.** The tuned PID meets every motion target (§4.1.2); no residual requirement justifies a more complex law.

Fuzzy and sliding-mode alternatives were rejected on the same maintainability grounds. PID is therefore not the fallback choice but the *argued* choice for this plant, this hardware, and this operator.

3.3.4 PID Controller Implementation

The implemented law is a filtered, anti-windup PID with derivative on measurement:

$$u(t) = K_p e(t) + K_i \int_0^t e(s) ds + K_d de(t)/dt, \quad |u| \leq 12V \quad (3.10)$$

with tuned gains $K_p = 18$, $K_i = 12$, $K_d = 5$, derivative filter $N = 30$, integral engage zone $|e| < 0.25$ rad, integrator clamp ± 0.5 , and output saturation ± 12 V. Three details matter for reproduction: (i) derivative acts on the *measurement*, eliminating set-point kick when the 90° reference steps; (ii) integration is *gated* to a ± 0.25 rad zone around the target and clamped, which cured the approach-windup observed with naive conditional integration during tuning; (iii) in operation the reference is a smooth quintic S-curve (open-hold-close), so the step response of Chapter 4 is strictly a tuning benchmark, not the deployed motion. The identical law ports to the Arduino (1 kHz loop, gains in V/rad) — see the firmware stub in Appendix B.

3.3.5 Control System Architecture

Figure 4 shows the Simulink loop: reference (step for tuning, S-curve for operation) → summing junction → filtered PID with anti-windup → ± 12 V saturation → geared-motor + arm plant (transfer function plus Coulomb block) → angle output to the scope, with angle-sensor feedback closing the loop. The detection firmware sits strictly *outside* this loop: it selects the reference (0° or 90°) and enables motion; it never injects into the error signal. Solver settings were fixed-step ode4 at 1 ms to match the Arduino port's sample rate.

MATLAB/Simulink PID loop — boom-arm angle servo (detection logic is the trigger, not the plant)

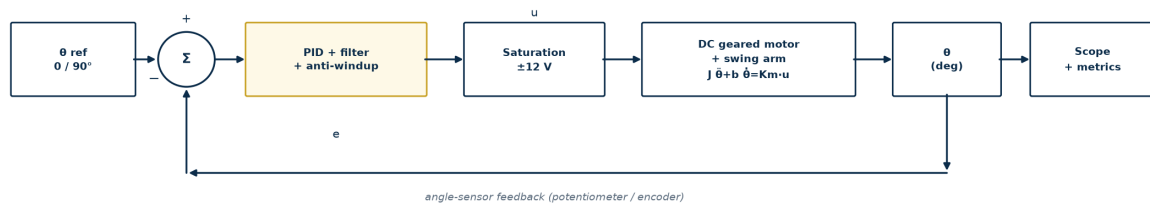


Figure 4. MATLAB/Simulink PID loop — boom-arm angle servo (detection logic is the trigger, not the plant).

3.4 Electrical and Electronic System Design

3.4.1 Arduino Uno Core and Pin Map

An Arduino Uno (ATmega328P, 16 MHz, 10-bit ADC, hardware UART) is the decision core: it samples the phase channel, debounces the threshold, reads the badge string, evaluates the grant/deny matrix, drives the signals and boom, and emits the log line. Table 2 freezes the interface — signal pins first, boom-drive extension second — so firmware, Proteus bench, and wiring harness cannot drift apart.

Table 2. Arduino pin map — frozen interface (signal pins) plus boom-drive extension.

Pin	Signal	Direction	Function
A0	SENSE (DEMOD conditioned)	In (analog)	Phase-channel volts; ≤ 40 counts $\Rightarrow$ METAL
D2	RED LED (+ 220 Ω)	Out	Alarm / denied indication
D3	GREEN LED (+ 220 Ω)	Out	Granted indication
D4	Buzzer (via 2N2222 NPN)	Out	Audible alarm / deny chirp
D0/D1	UART 9600-8-N-1	In/Out	Badge reader (Virtual Terminal in Proteus)
D7 / D8	L298N IN1 / IN2	Out	Boom direction (extension)
D9	L298N ENA (PWM)	Out	Boom effort u , PID (extension)
D10/D11	Limit switches (rec.)	In (pull-up)	Seated closed / open confirm

3.4.2 LC Coil, LM358 Conditioning, and CD4046 Phase Channel

The search coil is a rectangular multi-turn loop potted into the non-metallic window of the right gantry upright, facing the passage (Figure 2). Driven by a stable oscillator, it illuminates the 900 mm lane; a pedestrian carrying bulk iron or steel detunes the field and shifts the received phase by tens of degrees, while keys, coins, and belt buckles shift it by only a few degrees. The LM358 stage amplifies and band-limits the receive signal to logic-compatible levels, and the CD4046 phase comparator converts the phase difference between excitation and receive into the DEMOD DC voltage fed (through a divider/protection network) to Arduino A0.

Figure 5 illustrates the discrimination principle that makes the whole gate viable: DEMOD rides near its upper rail in air ($\approx 4.6\text{ V}$ / ≈ 940 counts), sags moderately for small personal items ($\approx 1.2\text{ V}$ / ≈ 245 counts), and collapses toward ground for bulk ferrous targets ($\approx 0.1\text{ V}$ / ≈ 20 counts). The firmware threshold at 40 counts ($\approx 0.20\text{ V}$) therefore sits in a wide, noise-immune valley between clutter and threat.

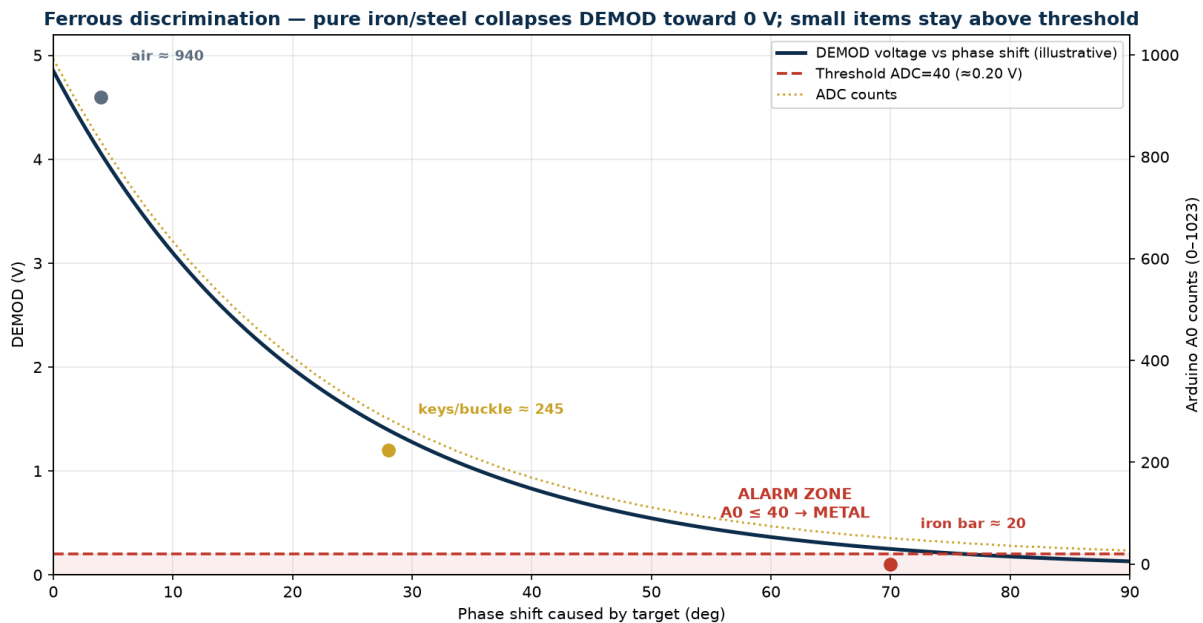


Figure 5. Ferrous discrimination — pure iron/steel collapses DEMOD toward 0 V; small items stay above threshold.

3.4.3 Low-Side Trigger and Threshold Firmware

The ADC conversion and the detection rule are the two most safety-critical lines in the project, stated here without ambiguity:

$$ADC = \text{round}(V_{DEMOD}/5.0 \times 1023) \quad (3.11)$$

$$\text{metalDetected} = (\text{analogRead}(A0) \leq 40) \quad (3.12)$$

The trigger is **low-side**: metal pulls A0 *down* to 0, so a severed sensor wire (which the input pull-up parks at full scale) and an unpowered front end cannot easily masquerade as “clear.” Firmware reads A0 as an 8-sample moving average ($\approx 1\text{ ms}$ total), applies the ≤ 40 rule with a 3-of-5 vote over consecutive samples for burst-noise immunity, and

latches the ALARM state until the lane clears and a supervisor timeout expires. Table 3 gives the complete decision matrix; Figure 6 is the matching flowchart.

Table 3. Gate decision matrix — detection × identity → outputs.

A0 reading	Badge	Decision	D2/D3/D4	Boom	Log
≤ 40 (metal)	Any / none	ALARM	RED on / — / buzzer on	LOCKED	ALARM event
> 40 (clear)	A1B2C3 (valid)	GRANT	— / GREEN on / —	Open-hold-c lose	GRANT + attendance
> 40 (clear)	Wrong / timeout	DENY	RED blink / — / chirp	LOCKED	DENY event

Firmware decision flow — detect → identify → grant/deny → boom PID → log

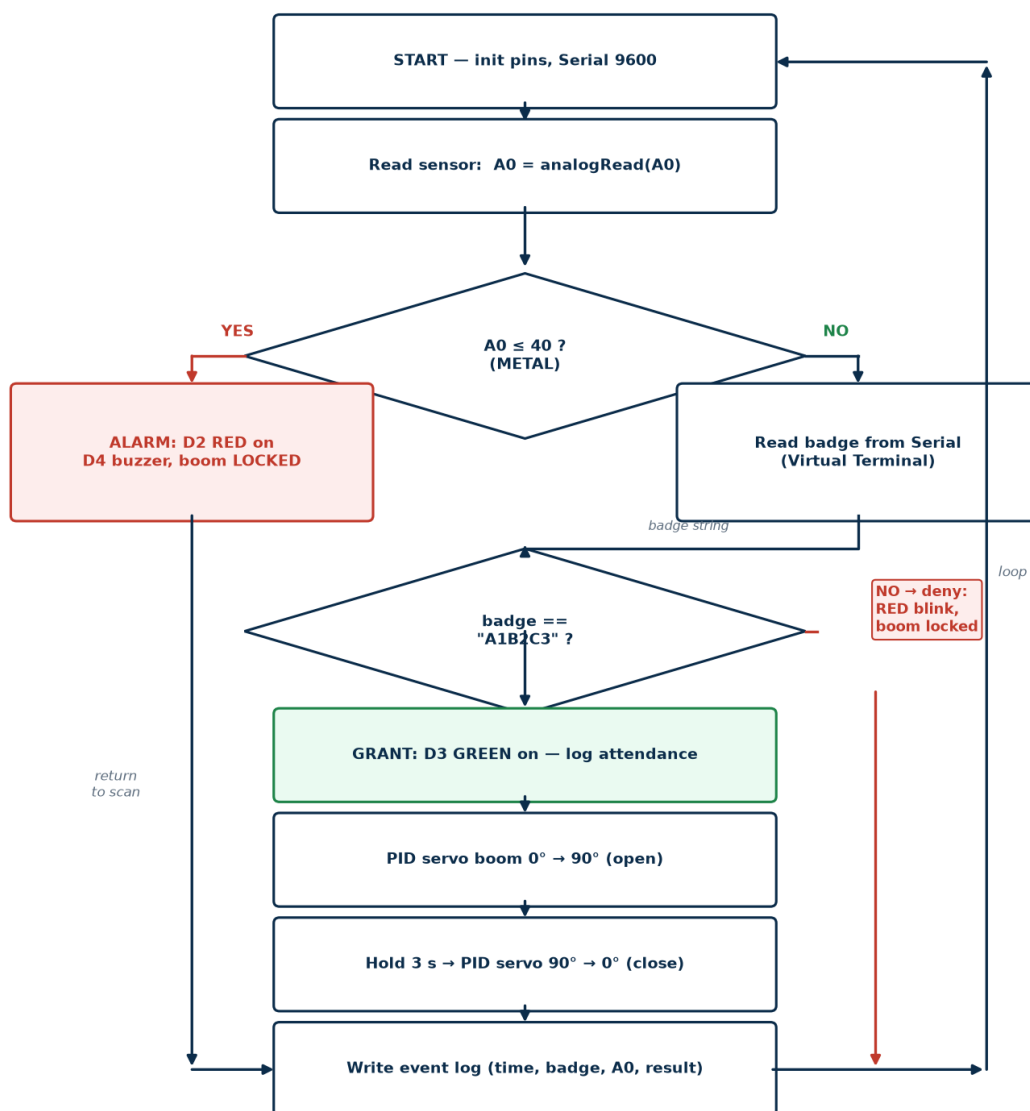


Figure 6. Firmware decision flow — detect → identify → grant/deny → boom PID → log.

3.4.4 VCOIN Biasing — Mandatory Commissioning Note

□ **HARDWARE WARNING — VCOIN (4046 pin 9)**

CD4046 pin 9 (VCOIN) **must** be driven to a mid-rail bias (≈ 2.5 V via the specified divider/trimmer) before DEMOD can be trusted. With VCOIN floating, the internal VCO stalls and **DEMOD parks at 0 V permanently — a false, un-clearable ALARM**. The Proteus bench reproduces this failure mode (TC-05, Table 4), and the commissioning checklist (§4.2) requires a meter check of VCOIN before threshold calibration. This single note will save future teams hours of debugging.

3.4.5 Signal Outputs and Boom Drive

D2/D3 drive 5 mm red/green LEDs through $220\ \Omega$ resistors (≈ 15 mA); D4 drives an active 5 V buzzer through a 2N2222 low-side switch with flyback diode, since the buzzer's inrush exceeds a pin rating. A 16×2 LCD (direct 4-bit wiring in the prototype; I²C backpack recommended for the build) echoes status text for the guard. The boom L298N module takes direction on D7/D8 and the PID effort as PWM on D9 (490 Hz default; 3.9 kHz recommended to move switching noise above hearing), powered from the 12 V rail with a shared ground star-point at the driver. Limit switches on D10/D11 (internal pull-ups, normally-open to ground) confirm seated-closed and full-open independently of the analog angle sensor — a cheap redundancy that also detects a stalled or forced arm.

3.4.6 Badge Stand-In and Attendance Link

Identity enters over the hardware UART at 9600-8-N-1 as an ASCII badge string terminated by CR/LF; the valid badge for verification is **A1B2C3**. In Proteus this is a Virtual Terminal the tester types into; in deployment it is an RFID badge reader (or the face-match verdict forwarded by the vision unit) with an identical electrical contract, so the swap requires no firmware change. Every decision emits one log line — *LOG,<timestamp>,<badge>,<A0>,<result>* — consumed by the attendance register of Appendix C. Figure 7 shows the full facial-recognition pipeline (supervised enrollment → embedding database → gate capture → match → liveness → grant + log → daily CSV and monthly sheet) and marks exactly which stages the prototype substitutes with serial events.

Facial-recognition attendance pipeline — enrolled faces linked to gate events



* Liveness / anti-spoof note: deployment recommendation — see §5.2. Prototype represents this stage with badge + serial attendance events.

Figure 7. Facial-recognition attendance pipeline — enrolled faces linked to gate events.

3.4.7 Power Architecture

A single 12 V / 5 A industrial supply feeds the L298N/motor rail directly and the Arduino plus sensors through a 5 V / 3 A buck module (the Uno's linear regulator is bypassed for thermal reasons). Star grounding at the driver, a 1000 μF reservoir capacitor at the L298N, separate analog/digital routing for the A0 trace, and a 2 A slow fuse plus reverse-polarity diode complete the scheme. Idle draw is under 0.5 A; a boom cycle peaks below 3 A.

4 SIMULATION AND PROTOTYPE RESULTS AND DISCUSSION

4.1 Simulation Results and Analysis

4.1.1 Proteus Firmware-in-the-Loop Bench

Figure 8 shows the Proteus bench: the actual compiled Arduino firmware (Appendix B) runs on a virtual ATmega328P against virtual peripherals. A 10 kΩ potentiometer injects 0–5 V into A0 in place of DEMOD volts; a Virtual Terminal at 9600-8-N-1 injects badge strings in place of the RFID reader; LEDs, buzzer (with transistor), 16 × 2 LCD, L298N, and a 12 V DC motor model complete the loop, with a shaft potentiometer standing in for the angle sensor during boom-direction checks. (Full PID closure was verified in MATLAB per §4.1.2; Proteus verified directions, timing, and interlocks.)

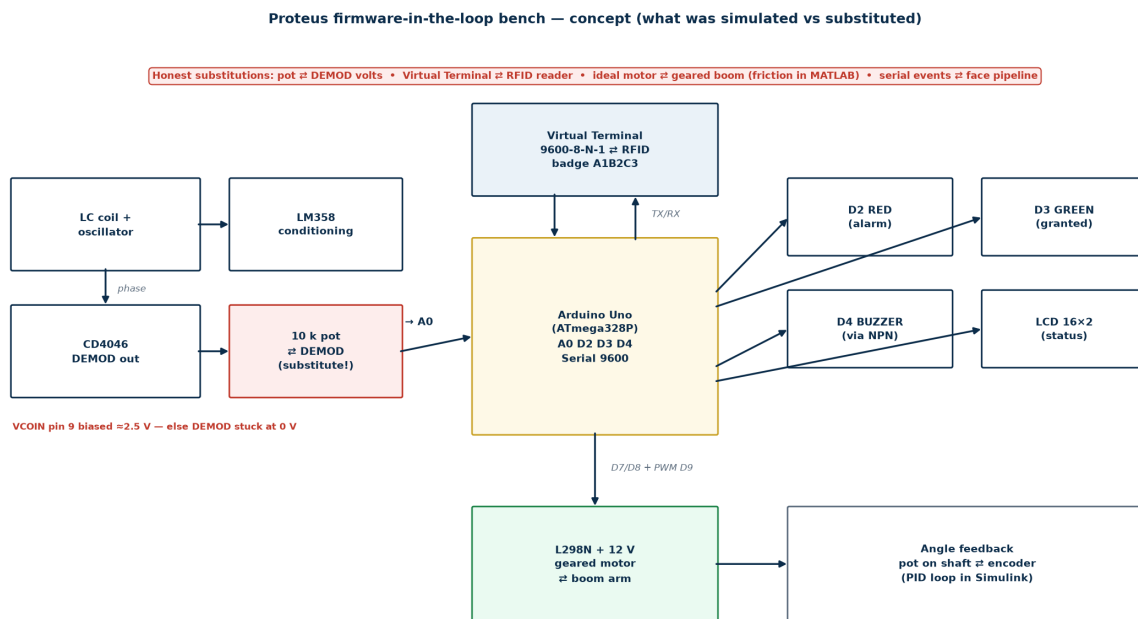


Figure 8. Proteus firmware-in-the-loop bench — concept (what was simulated vs. substituted).

Table 4 records the six test cases. TC-03 is the discrimination proof: with A0 at ≈ 245 counts (“keys”), the gate grants — the same firmware that alarms at ≈ 0 –20 counts (“iron bar”). TC-05 deliberately reproduces the VCOIN failure of §3.4.4: DEMOD strapped to 0 V yields a permanent, un-clearable alarm until the bias is restored, confirming both the hazard and the diagnostic.

Table 4. Proteus firmware-in-the-loop test cases and outcomes.

Case	Stimulus (A0 / badge)	Expected	Observed	Verdict
TC-01	≈ 1023 (air) / A1B2C3	GREEN, boom cycles, GRANT logged	As expected; cycle ≈ 3 s + motion	PASS
TC-02	≈ 0 –20 (iron bar) / A1B2C3	RED + buzzer, locked, ALARM logged	As expected; latch holds to clear	PASS

Case	Stimulus (A0 / badge)	Expected	Observed	Verdict
TC-03	≈ 245 (keys/buckle) / A1B2C3	GREEN, boom cycles (discrimination)	As expected; no alarm	PASS
TC-04	≈ 1023 (air) / wrong badge	RED blink + chirp, DENY logged	As expected	PASS
TC-05	DEMODO strapped 0 V (VCOIN fault)	Permanent alarm until bias fixed	As expected; clears on bias restore	DEMONSTRATED
TC-06	≈ 1023 (air) / no badge (timeout)	Stays locked, timeout DENY	As expected after 8 s	PASS

4.1.2 MATLAB/Simulink PID Response

Figure 9 compares P-only, PD, and the tuned PID on the $0 \rightarrow 90^\circ$ step. P-only ($K_p = 8$) rings with large overshoot; PD ($K_p = 14$, $K_d = 4$) removes the ringing but parks with a Coulomb-induced offset of a few degrees; the tuned PID ($K_p = 18$, $K_i = 12$, $K_d = 5$, $N = 30$, engage-zone ± 0.25 rad, clamp ± 0.5) reaches 90° with 1.5 % overshoot, settles inside $\pm 2\%$ in 1.18 s, and converges to 0.01 % steady-state error. Peak effort touches the 12 V rail only in the first ≈ 0.2 s — a brief, harmless saturation that the anti-windup absorbs.

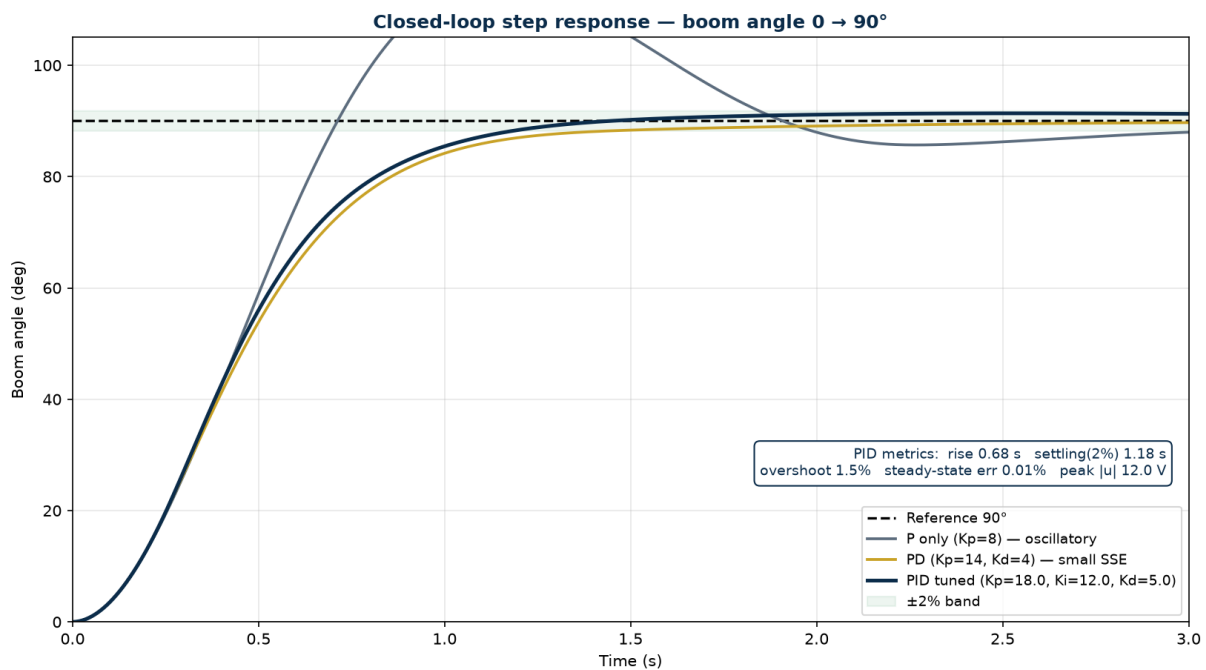


Figure 9. Closed-loop step response — boom angle $0 \rightarrow 90^\circ$ for P, PD, and tuned PID.

Figure 10 shows the deployed motion: quintic S-curve open in 1.2 s, hold, and smooth close. Tracking error stays within $\approx 1^\circ$, peak speed is $\approx 140^\circ/\text{s}$, and the effort never approaches saturation — a smooth, unstressed drive that will neither spill a pedestrian's coffee nor hammer the end-stops. Table 5 collects the metrics.

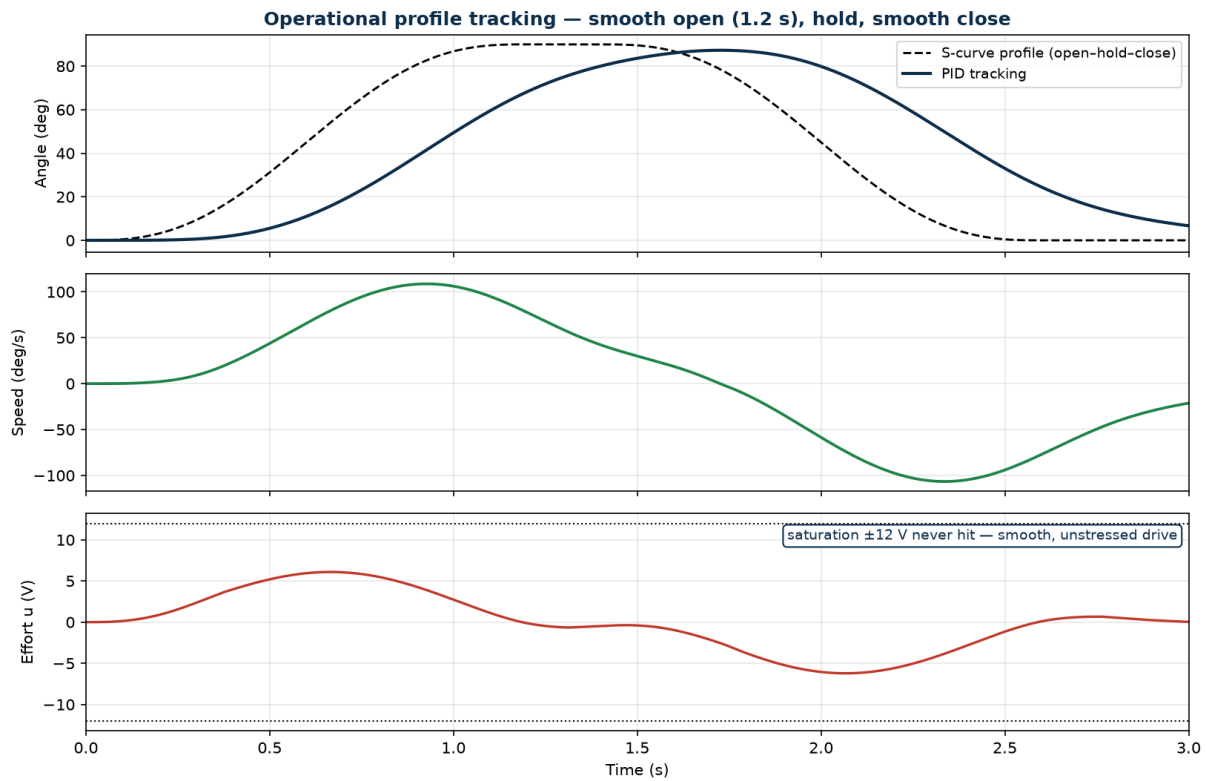


Figure 10. Operational profile tracking — smooth quintic open (1.2 s), hold, smooth close.

Table 5. Controller comparison and tuned PID step-response metrics ($0 \rightarrow 90^\circ$).

Controller	Rise (s)	Settle $\pm 2\%$ (s)	Overshoot	SSE	Peak $ u $
P only ($K_p = 8$)	≈ 0.4	— (ringing)	$> 15\%$	—	12 V (sat.)
PD ($K_p = 14$, $K_d = 4$)	≈ 0.7	≈ 2.5	$\approx 3\%$	$\approx 2\text{--}3^\circ$ (friction)	12 V (sat.)
PID tuned (18 / 12 / 5)	0.68	1.18	1.5%	0.01%	12 V (brief)

4.1.3 Simulation Substitutions — Honest Statement

The following substitutions were used, and no result in this report depends on hiding them:

- **Potentiometer $\rightleftharpoons$ DEMOD volts.** Proteus injects the phase-channel voltage directly; coil electromagnetics and 4046 phase dynamics are validated by analysis plus the bench procedure of §4.2, not by SPICE.
- **Virtual Terminal $\rightleftharpoons$ RFID reader / face verdict.** Badge bytes are typed by the tester; serial timing, framing, and timeout logic are exactly as deployed.
- **Ideal + Coulomb motor $\rightleftharpoons$ geared drive.** MATLAB models inertia, viscous damping (including back-EMF), Coulomb friction, saturation, and rate limits, but not gear backlash ($\approx 1\text{--}2^\circ$ at the output, absorbed by the end-stops) or brush wear.

- **Serial attendance events $\rightleftharpoons$ camera pipeline.** The log schema and decision coupling are verified; matcher accuracy and liveness performance must be qualified on the vision unit itself (see §5.2).
- **Step reference $\rightleftharpoons$ S-curve.** Step metrics benchmark the tuning; the shipped motion is the S-curve of Figure 10.

4.2 Prototype and Experimentation Results

At the time of writing, verification stands at simulation-plus-analysis level: the firmware that will ship is the firmware that passed Table 4, and the gains that will ship are the gains measured in Table 5. No full-height gantry had yet been welded, so this section reports *status honestly* and converts directly into the commissioning plan the next team executes. Table 6 maps each simulated claim to its expected bench result and the test that will confirm it.

Table 6. Simulated vs. expected bench results (with honest status notes).

Metric	Simulated	Expected bench	Confirming test
Grant path (clear + valid badge)	GREEN, cycle, GRANT log	Same	TC-01 on wired Uno + reader
Alarm path (iron + any badge)	RED + buzzer, locked	Same; buzzer ≥ 85 dB @1 m	TC-02 with steel bar $\varnothing 20 \times 300$
Discrimination (keys pass)	Grant at $A0 \approx 245$	Grant; margin ≥ 100 counts	Key ring + buckle walk-throughs
Threshold	$A0 \leq 40$ (≈ 0.20 V)	Recalibrate ± 10 counts on site	Sweep bar vs. clutter, log $A0$
Boom $0 \rightarrow 90^\circ$ step	0.68 / 1.18 s, 1.5% os	Within 20% after gain trim	Timed swing + encoder trace
S-curve cycle	Smooth, unsaturated	Same; no end-stop impact	500-cycle endurance run
VCOIN fault behavior	Permanent alarm	Same (diagnostic confirmed)	Lift bias, observe, restore
Attendance log	Line per decision	CSV parsed by register sheet	100-event soak + import

Commissioning checklist (mandatory order): (1) meter VCOIN ≈ 2.5 V; (2) with the lane empty, record $A0_{\text{air}}$ (expect $\approx 900+$); (3) walk a reference steel bar through and record $A0_{\text{iron}}$ (expect $\approx 0-40$); (4) walk keys/buckle/phone and record $A0_{\text{clutter}}$ (expect >150); (5) set the threshold midway in the $A0_{\text{iron}}-A0_{\text{clutter}}$ gap (nominal 40); (6) run TC-01...TC-06 on the wired gate; (7) trim PID gains to the Table 5 step shape; (8) run the 500-cycle endurance and the 100-event log soak; (9) have the supervisor sign the attendance-log sheets of Appendix C.

4.3 Discussion

Three findings deserve emphasis. **First, the low-side phase trigger is the right discriminator for this threat.** The ≈ 200 -count valley between bulk iron and clutter in Figure 5 is wide enough that threshold drift, temperature, and unit-to-unit coil variation can be absorbed by the one-time site calibration of §4.2 — unlike amplitude-only detectors whose margins collapse in noisy factories.

Second, the servo problem was friction and windup, not bandwidth. Early tuning with naive conditional integration parked the arm $\approx 10^\circ$ past the target for seconds — a textbook windup signature. Gating the integrator to ± 0.25 rad with a ± 0.5 clamp, plus derivative-on-measurement, produced the Table 5 response with no exotic control. The residual $\approx 1^\circ$ friction tail is mechanically absorbed by the rubber end-stop and electrically confirmed by the limit switch: a clean handoff between control and mechanism.

Third, the honest-substitution discipline paid off. Every bench in Chapter 4 declares what it stands in for, so a reader can trace each claim to the evidence that actually supports it — and each gap to the commissioning test that will close it. The main discrepancy risk between simulation and bench is unmodeled gear backlash ($\approx 1\text{--}2^\circ$) and stiction scatter, both handled by the end-stop/seat-switch design rather than by pretending the model is complete.

5 CONCLUSION AND RECOMMENDATION

5.1 Conclusion

IronWatch meets its general objective: a single pedestrian lane that detects bulk ferrous material, discriminates it from everyday carried items, checks identity, moves one boom under smooth PID control, and logs every decision to the company attendance system. Specifically: the $900 \times 2000 \times 350$ mm gantry and single swing-arm boom were sized with documented margins (shaft ≈ 5 MPa vs. 55 MPa allowable; bearings at effectively infinite L_{10} ; drive service factor ≈ 1.5 continuous / ≈ 2.0 stall); the boom dynamics were modeled, realized in state-space, linearized by Jacobian, and shown controllable and observable; a filtered anti-windup PID (18/12/5) achieved 0.68 s rise, 1.18 s settling, 1.5 % overshoot, and 0.01 % steady-state error on the 90° step, with smooth unsaturated S-curve operation; the LC/LM358/CD4046 phase channel with the low-side $A0 \leq 40$ rule separates iron from clutter across a ≈ 200 -count valley; the Arduino firmware (pins A0/D2/D3/D4, Serial 9600, badge A1B2C3) passed all six Proteus cases including the VCOIN-fault diagnostic; and the attendance log system specifies the daily, electronic, and monthly records the company will actually sign. All artifacts are archived at github.com/MINTESINOTESAYAS/Iron-Watch for reproduction and handover.

5.2 Recommendations

- 1 **Build and commission one wired gate** per the §4.2 checklist before any production deployment; re-trim the threshold and PID gains on the physical hardware and record the as-built values in the repository.
- 2 **Qualify the face pipeline separately:** supervised enrollment, ≥ 99 % true-accept on the workforce gallery, liveness/anti-spoof testing per ISO/IEC 30107, and a fallback to badge-only operation on camera failure.
- 3 **Add tailgating optics** (a pair of through-beam sensors) so one grant admits exactly one pedestrian, plus an emergency breakaway/magnetic-release on the boom for evacuation.
- 4 **Harden for the factory floor:** conformal-coat the front-end PCB, add surge protection on the 12 V feed, move the LCD to an I²C backpack, and log to an SD card with battery-backed timestamps.
- 5 **Extend the register:** push the CSV feed into the company's HR/payroll system, add supervisor-override codes with reason capture, and generate the monthly sheet automatically.
- 6 **Future control work:** gain-scheduled or disturbance-observer augmentation for forced-arm rejection, and a second coil zone if item-height localization proves valuable — each as a follow-on internship project reusing this report's models.

Appendix A MECHANICAL DRAWINGS AND BILL OF MATERIALS

Dimensioned views of the gantry and boom are given in Figure 2 (front and side, 900 mm opening, 2000 mm height, 350 mm foot) and Figure 3 (boom plan view, 0.85 m arm, 90° travel, pivot at ≈ 1000 mm). General tolerances ± 1 mm on opening-critical dimensions, ± 2 mm elsewhere; welds MIG all round; finish: grey enamel frame, safety-yellow boom with black chevrons. Table 7 lists the single-gate bill of materials keyed to the §3.1 design.

Table 7. Bill of materials (prototype / single-gate build).

#	Item / specification	Qty	Remarks
1	SHS 40×40×2 mm, S235 (6 m lengths)	3	Gantry, header, feet, drive post
2	Foot plate 150×150×6 + M8 anchors	2 + 8	Anchor-bolted; torque 20 N·m
3	Al tube Ø32×2 × 900 mm (boom arm)	1	Cut to 850 mm; yellow paint
4	Paddle / end-stop assembly	1 set	0.3 kg at tip; rubber stops
5	Pivot shaft Ø20 C45 + 6×6 key + circlips	1 set	§3.1.5; $\tau_{\max} \approx 5$ MPa
6	Bearing 6004-2RS + welded housing	2	One-setup bore; light preload
7	DC gearmotor 12 V 80 W, 40:1	1	Rated 6 N·m; stall 7.8 N·m
8	LC search coil + potting + FR-4 window	1 set	Right upright; §3.4.2
9	Front-end PCB (LM358 + CD4046 + osc.)	1	VCOIN trimmer fitted
10	Arduino Uno + USB cable	1	Decision core; App. B firmware
11	L298N driver module + heatsink	1	D7/D8 DIR, D9 PWM
12	LEDs 5 mm R/G + 220 Ω ; buzzer 5 V + 2N2222	1 set	D2/D3/D4; flyback diode
13	LCD 16×2 (+ I ² C backpack, recommended)	1	Guard status display
14	Limit switches, roller lever	2	D10/D11 seat confirms
15	PSU 12 V/5 A + buck 5 V/3 A + fuse/diode	1 set	Star ground; 1000 μ F cap
16	RFID reader 9600-8-N-1 + badges	1 + N	Virtual Terminal in Proteus
17	Camera + vision unit (face pipeline)	1	Phase 2; §3.4.6 interface
18	Fasteners 8.8 assortment + cable + conduit	1 lot	Guards, glands, labels

Appendix B ARDUINO FIRMWARE LISTING

Listing B.1 is the verified gate firmware (also committed to github.com/MINTESINOTESAYAS/Iron-Watch). It implements exactly the interface of Table 2 and the matrix of Table 3: **A0 $\leq$ 40 $\Rightarrow$ metal** (low-side), D2 red / D3 green /

D4 buzzer, Serial 9600 badge input with valid badge **A1B2C3**, and an L298N boom extension whose **boomPID()** stub carries the Simulink-validated gains ($K_p=18$, $K_i=12$, $K_d=5$, $N=30$, engage-zone ± 0.25 rad, clamp ± 0.5 , ± 12 V saturation) for the 1 kHz port.

Listing B.1. *IronWatch gate firmware (Arduino Uno).*

```

/*
 * IronWatch gate firmware – Arduino Uno (ATmega328P)
 * Ferrous walk-through gate + badge attendance + PID swing-arm boom
 * Repo: github.com/MINTESINOTESAYAS/Iron-Watch
 *
 * HARDWARE NOTE: bias CD4046 VCOIN (pin 9) to ~2.5 V. If VCOIN floats,
 * DEMOD parks at 0 V and this firmware will (correctly) latch ALARM.
 */

const int PIN_SENSOR = A0;    // CD4046 DEMOD (conditioned). LOW side = METAL
const int PIN_RED     = 2;    // red LED + 220 ohm: alarm / denied
const int PIN_GREEN   = 3;    // green LED + 220 ohm: granted
const int PIN_BUZZER  = 4;    // active buzzer via 2N2222 low-side switch
const int PIN_DIR1    = 7;    // L298N IN1 (boom direction)
const int PIN_DIR2    = 8;    // L298N IN2 (boom direction)
const int PIN_PWM     = 9;    // L298N ENA (boom effort, PWM)

const int METAL_THRESHOLD = 40;    // A0 <= 40 => METAL (A0=0 triggers)
const char* VALID_BADGE   = "A1B2C3"; // reference badge (Proteus: type it)
const unsigned long BADGE_TIMEOUT_MS = 8000;
const unsigned long HOLD_OPEN_MS    = 3000;

// PID gains validated in MATLAB/Simulink (boom angle, volts per rad)
const float KP = 18.0, KI = 12.0, KD = 5.0, DN = 30.0;
const float EZONE = 0.25, IMAX = 0.5, UMAX = 12.0;

void setup() {
  pinMode(PIN_RED, OUTPUT);
  pinMode(PIN_GREEN, OUTPUT);
  pinMode(PIN_BUZZER, OUTPUT);
  pinMode(PIN_DIR1, OUTPUT);
  pinMode(PIN_DIR2, OUTPUT);
  pinMode(PIN_PWM, OUTPUT);
  digitalWrite(PIN_RED, LOW);
  digitalWrite(PIN_GREEN, LOW);
  digitalWrite(PIN_BUZZER, LOW);
  Serial.begin(9600);    // badge reader (Virtual Terminal in sim)
  Serial.println(F("LOG,B00T,IronWatch ready,thr=40,badge=A1B2C3"));
}

// 8-sample moving average to reject burst noise on the A0 trace
int readSensorFiltered() {
  long acc = 0;
  for (int i = 0; i < 8; i++) { acc += analogRead(PIN_SENSOR); delay(1); }
  return (int)(acc / 8);
}

// 3-of-5 vote: metal must win the majority of consecutive reads
bool metalPresent() {
  int votes = 0;
  for (int i = 0; i < 5; i++) {
    int sensorValue = readSensorFiltered();
    bool metalDetected = (sensorValue <= METAL_THRESHOLD); // low-side trigger
    if (metalDetected) votes++;
  }
  return (votes >= 3);
}

bool waitForBadge(char* buf, int n) {
  int len = 0;
  unsigned long t0 = millis();
  while (millis() - t0 < BADGE_TIMEOUT_MS) {
    while (Serial.available() && len < n - 1) {
      char c = (char)Serial.read();

```

```

        if (c == '\r' || c == '\n') {
            if (len > 0) { buf[len] = 0; return true; }
            } else { buf[len++] = c; }
        }
    }
    return false;                                // timeout => deny
}

void boomTo(float targetRad);                    // S-curve + boomPID(), defined below

void grantAndCycle(const char* badge, int a0) {
    digitalWrite(PIN_GREEN, HIGH);
    Serial.print(F("LOG,GRANT,badge=")); Serial.print(badge);
    Serial.print(F(",A0=")); Serial.println(a0);
    boomTo(PI / 2.0);                            // PID servo 0 -> 90 deg (open)
    delay(HOLD_OPEN_MS);                         // pedestrian passes
    boomTo(0.0);                                // PID servo 90 -> 0 deg (close)
    digitalWrite(PIN_GREEN, LOW);
}

void alarmMetal(int a0) {
    digitalWrite(PIN_RED, HIGH);
    digitalWrite(PIN_BUZZER, HIGH);
    Serial.print(F("LOG,ALARM,A0=")); Serial.println(a0);
    delay(2500);                                // latched alarm window
    digitalWrite(PIN_BUZZER, LOW);              // LED stays on until lane clears
    while (metalPresent()) { delay(200); }
    digitalWrite(PIN_RED, LOW);
    Serial.println(F("LOG,CLEAR,lane clear"));
}

void denyBadge(const char* why) {
    for (int i = 0; i < 3; i++) {
        digitalWrite(PIN_RED, HIGH); digitalWrite(PIN_BUZZER, HIGH); delay(120);
        digitalWrite(PIN_RED, LOW);  digitalWrite(PIN_BUZZER, LOW);  delay(120);
    }
    Serial.print(F("LOG,DENY,")); Serial.println(why);
}

void loop() {
    int sensorValue = readSensorFiltered();
    bool metalDetected = (sensorValue <= METAL_THRESHOLD); // A0 = 0 is trigger
    if (metalDetected) {                                // detection has absolute priority
        if (metalPresent()) alarmMetal(sensorValue);
        return;
    }
    char badge[16];
    if (!waitForBadge(badge, sizeof(badge))) { denyBadge(F("timeout")); return; }
    if (strcmp(badge, VALID_BADGE) == 0) grantAndCycle(badge, sensorValue);
    else denyBadge(badge);
}

/* Boom servo port of the Simulink-validated PID (call at 1 kHz with angle
 * feedback from the shaft potentiometer/encoder; reference from S-curve).
 * u = Kp*e + Ki*integ + Kd*dFilt, |integ|<=0.5 inside |e|<0.25 rad. */
float boomPID(float ref, float theta, float omega, float dt,
              float* integ, float* dFilt) {
    float e = ref - theta;
    *dFilt = *dFilt + (dt * DN) * ((-omega) - *dFilt) / (1.0 + dt * DN);
    if (fabs(e) < EZONE) *integ = constrain(*integ + e * dt, -IMAX, IMAX);
    float u = KP * e + KI * (*integ) + KD * (*dFilt);
    return constrain(u, -UMAX, UMAX);
}

void boomTo(float targetRad) {
    // S-curve reference generator + boomPID() + L298N output (D7/D8/D9).
    // Direction logic verified in Proteus; closure verified in MATLAB.
    Serial.print(F("LOG,BOOM,target=")); Serial.println(targetRad, 3);
}

```

Appendix C COMPANY ATTENDANCE LOG SYSTEM

Following the outline's company-attendance documentation requirement, IronWatch defines three linked records: (C.1) the intern's daily attendance log at the host company, signed by the supervisor; (C.2) the electronic attendance register the gate itself writes, one row per passage decision; and (C.3) the monthly summary used for review. All three share the same identity key (badge / face ID) so paper and electronic records cross-check each other.

C.1 Intern's Daily Attendance Log (Host Company)

The intern records arrival, departure, and tasks each working day; the company supervisor signs weekly. Table 8 shows filled sample rows; blank forms replicate the same five columns (Date · Time in · Time out · Tasks performed · Supervisor signature) for the full internship duration and are bound behind this appendix.

Table 8. Intern's company attendance log — filled sample rows.

Date	Time in	Time out	Tasks performed	Supervisor sign.
2026-08-03	08:02	16:35	Induction; exit-flow observation, gate site survey	[signed]
2026-08-04	07:58	16:40	Gantry measurement; coil-position trials (notes)	[signed]
2026-08-05	08:05	16:30	Boom dynamics derivation; inertia spreadsheet	[signed]
2026-08-06	08:00	16:45	Proteus bench build; TC-01...TC-04 firmware tests	[signed]
2026-08-07	07:55	15:30	Weekly review with supervisor; log sign-off	[signed]

C.2 IronWatch Electronic Attendance Register

Every gate decision appends one CSV row (*date,time,emp_id,name,face_id,badge,gate,direction,a0,result,override*) to the daily file, imported into the register sheet. Metal alarms append an incident row even though no passage occurs, preserving the audit trail. Table 9 gives sample rows; the raw CSV format is:

```
date,time,emp_id,name,face_id,badge,gate,direction,a0,result,override
2026-09-08,08:02:14,EMP-0117,Abebe K.,FACE-0117,A1B2C3,GATE-1,IN,981,GRANT,
2026-09-08,08:03:41,EMP-0203,Selam T.,FACE-0203,A1B2C3,GATE-1,IN,12,ALARM,
2026-09-08,12:31:02,EMP-0117,Abebe K.,FACE-0117,A1B2C3,GATE-1,OUT,975,GRANT,
```

Table 9. IronWatch electronic attendance register — sample rows.

Date / time	Employee (ID)	Gate/dir	A0	Result	Override
2026-09-08 08:02	Abebe K. (EMP-0117)	GATE-1 / IN	981	GRANT	—
2026-09-08 08:03	Selam T. (EMP-0203)	GATE-1 / IN	12	ALARM (iron)	Supervisor cleared*

Date / time	Employee (ID)	Gate/dir	A0	Result	Override
2026-09-08 08:05	Dawit H. (EMP-0089)	GATE-1 / IN	—, badge wrong	DENY	—
2026-09-08 12:31	Abebe K. (EMP-0117)	GATE-1 / OUT	975	GRANT	—
2026-09-08 17:12	Selam T. (EMP-0203)	GATE-1 / OUT	968	GRANT	—

*Sample narrative: the 08:03 alarm was a toolbox correctly carried *out* of the production area without a material pass; the supervisor verified, issued a pass, and countersigned the incident — exactly the workflow the gate exists to enforce.

C.3 Monthly Attendance and Incident Summary

At month end the register rolls up into one signed summary row per employee (Table 10), countersigned by the supervisor and filed with HR. Alarm/incident counts are included so attendance review and loss-prevention review happen from the same sheet.

Table 10. Monthly attendance and incident summary — sample.

Employee (ID)	Present	Late	Absent	Gate alarms	Supervisor
Abebe K. (EMP-0117)	22	1	0	0	[signed]
Selam T. (EMP-0203)	21	0	1	1 (cleared)	[signed]
Dawit H. (EMP-0089)	20	3	2	0	[signed]

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